

Android & iOS Native Screen Recorder

[Check online Documentation](#)

Record gameplay on both Android and iOS platforms with no extra effort. On Android it uses the native **MediaRecorder API** with a **Foreground Service**. On iOS it uses **ReplayKit**.

- **Android: Lollipop 5.1 and above (including Android 13+).**
- **iOS: iPhone, iPad and iPod Touch on iOS 9.0 and above.**

Setup

1. Add the prefab

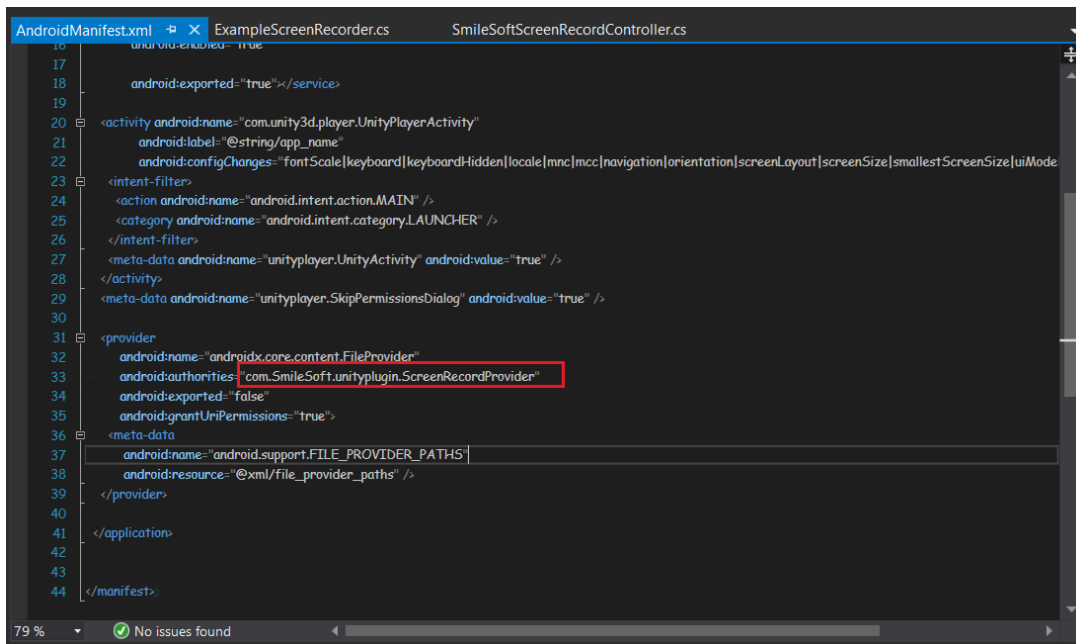
Drag the Screen Recorder prefab into your scene hierarchy from:

Assets > SunShine Android Native Screen Recorder > Prefab > Screen Recorder

2. Configure the file provider (Android only)

Only needed if you want to share videos through the native share dialog.

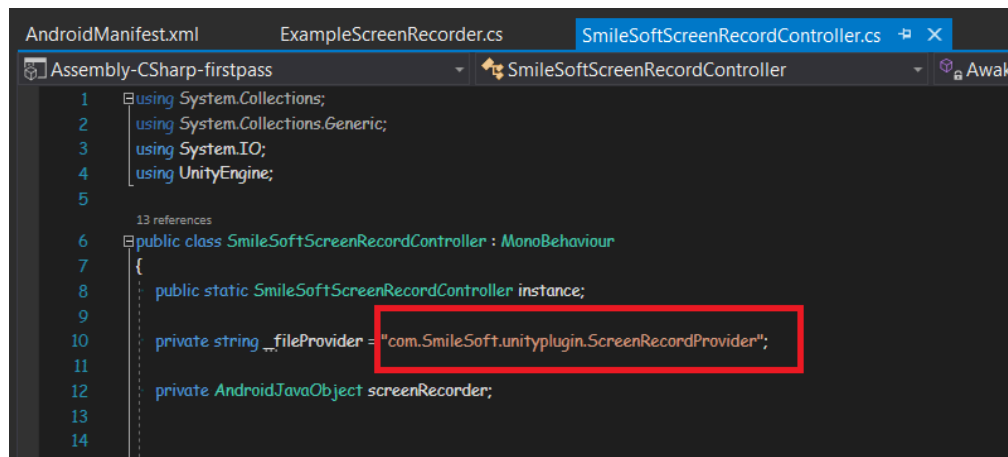
1. Open **Assets > Plugins > Android > AndroidManifest.xml** and set **android:authorities** to a unique value. Using your package name is recommended.



```
16 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
17     android:exported="true">
18     <service
19         android:name="com.smilesoft.screenrecord.ScreenRecorderService"
20         android:exported="true">
21         <intent-filter>
22             <action android:name="android.intent.action.MAIN" />
23             <category android:name="android.intent.category.LAUNCHER" />
24         </intent-filter>
25         <meta-data android:name="unityplayer.UnityActivity" android:value="true" />
26     </service>
27     <activity
28         android:name="com.unity3d.player.UnityPlayerActivity"
29         android:label="@string/app_name"
30         android:configChanges="fontScale|keyboard|keyboardHidden|locale|mnc|mcc|navigation|orientation|screenLayout|screenSize|smallestScreenSize|uiMode"
31         <intent-filter>
32             <action android:name="android.intent.action.MAIN" />
33             <category android:name="android.intent.category.LAUNCHER" />
34         </intent-filter>
35         <meta-data android:name="unityplayer.UnityActivity" android:value="true" />
36         <meta-data android:name="unityplayer.SkipPermissionsDialog" android:value="true" />
37     </activity>
38     <provider
39         android:name="androidx.core.content.FileProvider"
40         android:authorities="com.smilesoft.screenrecord.plugin.ScreenRecordProvider"
41         android:exported="false"
42         android:grantUriPermissions="true">
43         <meta-data
44             android:name="android.support.FILE_PROVIDER_PATHS"
45             android:resource="@xml/file_provider_paths" />
46     </provider>
47 </manifest>
```

AndroidManifest.xml — set android:authorities to a unique value.

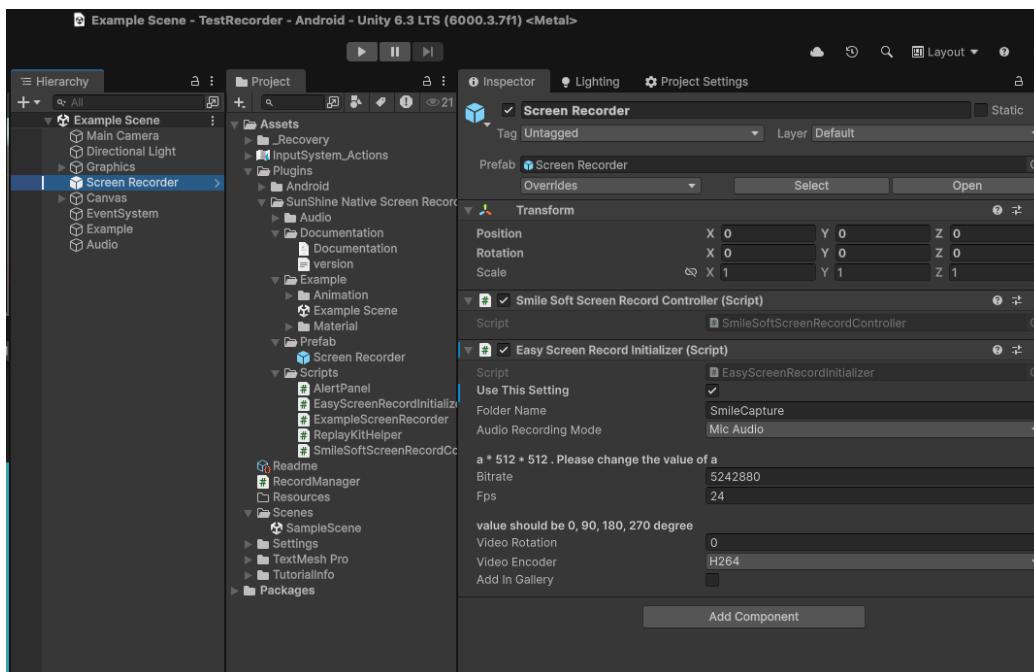
2. Open **SunShine Android Native Screen Recorder > Scripts > SmileSoftScreenRecordController.cs** and set the **_fileProvider** field to the same value you used in the manifest.



SmileSoftScreenRecordController.cs — set _fileProvider to match.

Easy Initializer

The Screen Recorder prefab includes a **ready-to-use initializer**. Toggle **Use This Setting** in the Inspector to enable it, then adjust the recorder settings (folder name, bitrate, FPS, resolution, encoder, rotation) directly from there.



The Easy Screen Record Initializer in the Inspector.

API Reference

All methods are called on **SmileSoftScreenRecordController.instance**. The tables below omit that prefix for readability.

Recording

Method	Description
<code>StartRecording()</code>	Starts recording. (Free version stops automatically after 5 seconds.)
<code>StopRecording()</code>	Stops recording. On Android, returns the recorded file path.

Audio

Choose the audio source before recording. On Android you can pick microphone or system audio; on iOS only the microphone is supported.

Method	Description
<code>SetAudioRecordingMode(mode)</code>	Sets the audio source using the <code>AudioRecordingMode</code> enum.

`AudioRecordingMode` values:

- **NoAudio** — record without audio.
- **SystemAudio** — in-game / system audio (Android only).
- **MicAudio** — microphone (Android & iOS).

Files & Storage (Android only)

Method	Description
<code>SetVideoDestination(destination)</code>	Sets the save location (string). Defaults to external storage.
<code>SetVideoName(fileName)</code>	Sets the output file name (string).
<code>SetStoredFolderName(folderName)</code>	Saves the video into a new folder (string).
<code>SetGalleryAddingCapabilities(bool)</code>	If true, adds the video to the device gallery. Default: true.

Save Options (iOS only)

Method	Description
<code>SetIosSaveToPhotos(bool)</code>	If true, saves the video to Photos. Default: true.
<code>SetIosSaveToDocuments(bool)</code>	If true, saves to the app's private directory. Default: true.

Video Quality (Android only)

Method	Description
<code>SetBitRate(bitrate)</code>	Sets bitrate (int). Higher = better quality. Tip: use $a \times 512 \times 512$ and change a.
<code>SetVideoSize(width, height)</code>	Sets resolution (both int).
<code>SetVideoEncoder(encoder)</code>	Sets the encoder (int). Check Android docs for support per API level.
<code>SetVideoRotation(rotation)</code>	Sets rotation (int): 0, 90, 180 or 270. Support varies by encoder/API.

Preview & Share

Method	Description
<code>PreviewVideo(filePath)</code>	Opens the native video preview. On iOS the path is not required.
<code>ShareVideo(filePath, message, title)</code>	Opens the native share dialog (Android). On iOS only filePath and message are used.

Note

The Audio, Files & Storage, Save Options and Video Quality methods are optional – use any or none. Just make sure you call them before `StartRecording()`.

Callbacks

These are received in `SmileSoftScreenRecordController.cs`.

Android

Callback	When it fires
<code>OnFreeVersionLimitReached</code>	Free version only: fires when recording passes 5 seconds. Returns JSON with videoPath and isAutoStoppedForFreeUser.
<code>OnRecordPermissionGranted</code>	Fires when the user accepts or rejects the permission popup. Status: "True" = accepted, "False" = declined.
<code>OnRecordStartStatus</code>	Fires when recording is attempted. Status: "True" = started, "False" = failed.

iOS

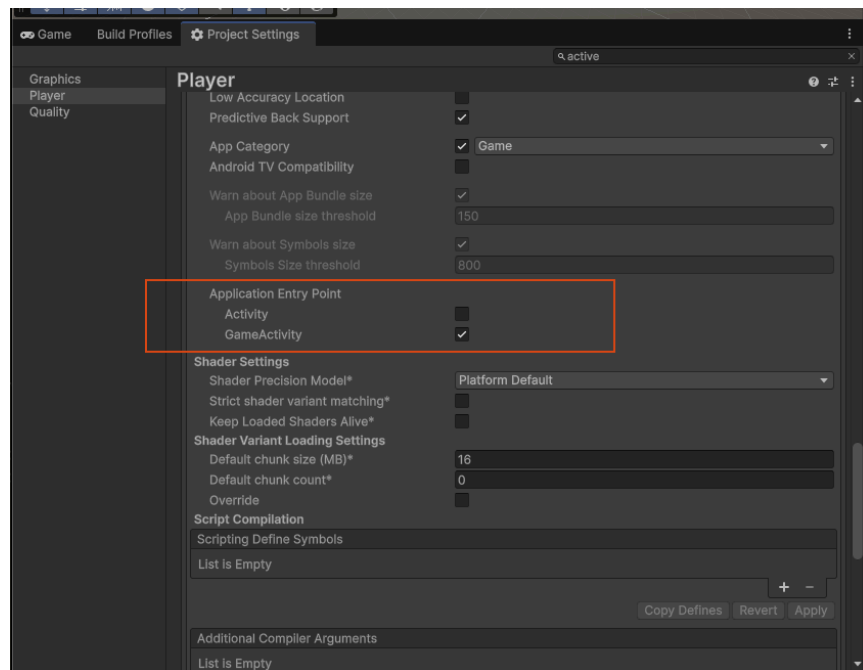
Callback	When it fires
<code>OnFreeVersionLimitReached</code>	Same as Android.

Callback	When it fires
OnIosRecordStatus	Fires when iOS recording starts or fails. Status (Boolean): true = started, false = failed.

Troubleshooting

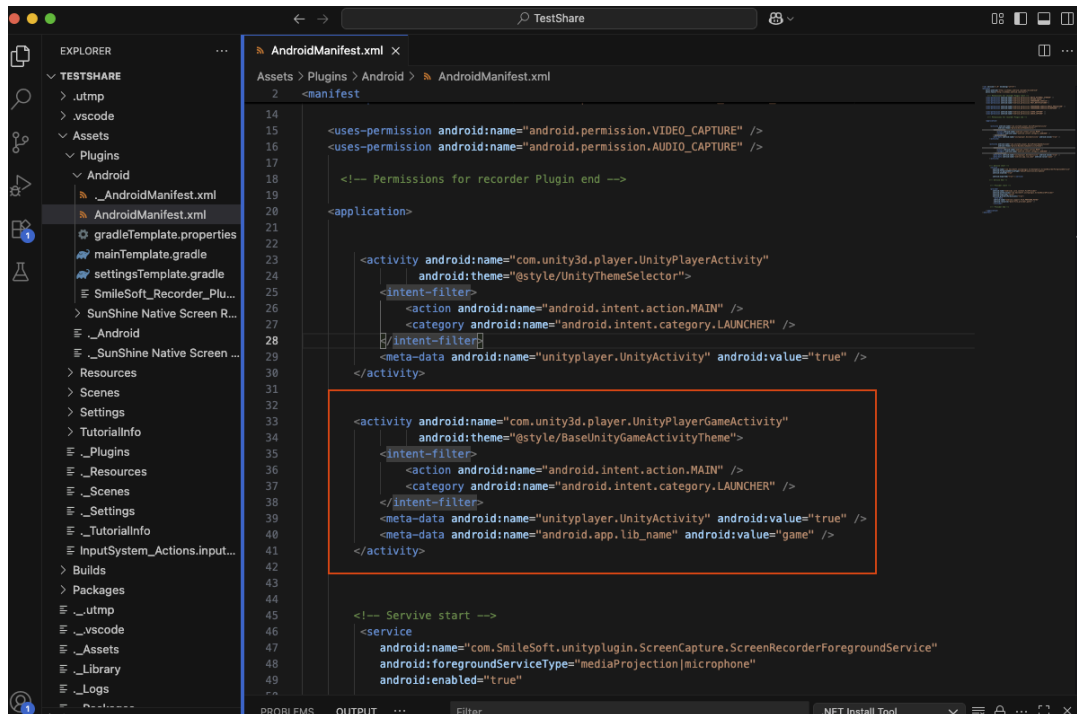
Android

- Go to Project Settings > Android > Other Settings > Application Entry Point and make sure Game Activity is selected.



Player settings — make sure Game Activity is the entry point.

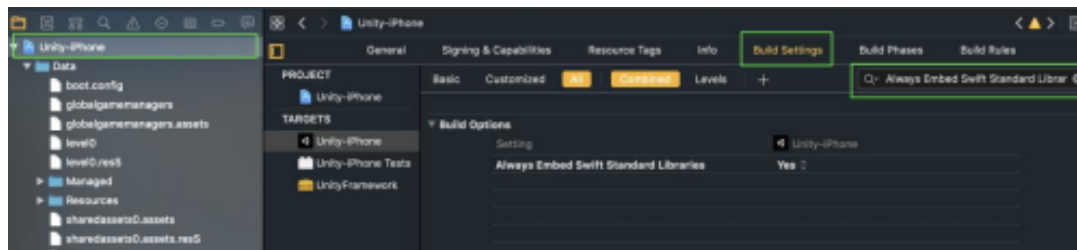
- If you don't want Game Activity as the entry point, remove the `UnityPlayerGameActivity` section from the activity block in the manifest.



The UnityPlayerGameActivity section to remove (highlighted).

ios

- Only filePath and message are used for sharing; everything else is set automatically.
- If you hit a **dyld: Library not loaded: @rpath/libswiftCore.dylib** error when building, open Xcode and set Always Embed Swift Standard Libraries to Yes under the Build Settings tab.



Xcode — set Always Embed Swift Standard Libraries to Yes.

Tip: For a complete working example, open the example scene at **Assets > SunShine Android Native Screen Recorder > Example > Example Scene**.